

Solution

Magnetic dipole is kept, the total flux emanating from the closed surface S is zero. So,

$$\Phi_B = \oint \vec{B}.\mathrm{d}\vec{A} = 0$$

Here the integral is taken over closed surface. Since no isolated magnetic pole (called magnetic monopole) exists, this integral is always zero,

$$\oint \vec{B}.\mathrm{d}\vec{A} = 0$$

This is similar to Gauss's law in electrostatics. (Refer unit 1)

3.2 COULOMB'S INVERSE SQUARE LAW OF MAGNETISM

Consider two bar magnets A and B as shown in Figure 3.14.

When the north pole of magnet A and the north pole of magnet B or the south pole of magnet A and the south pole of magnet B are brought closer, they repel each other. On the other hand, when the north pole of magnet A and the south pole of magnet B or the south pole of magnet A and the north pole of magnet B are brought closer, their poles attract each other. This looks similar to Coulomb's law for static charges studied

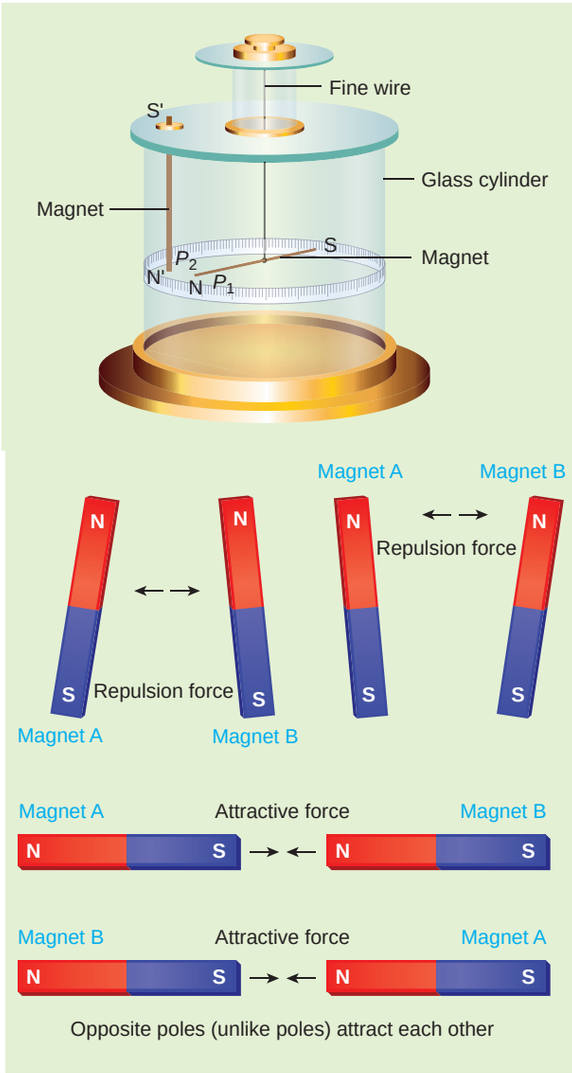


Figure 3.14: Magnetic poles behave like electric charges – like poles repel and unlike poles attract

in Unit I (opposite charges attract and like charges repel each other). So analogous to Coulomb's law in electrostatics, (Refer unit 1) we can state Coulomb's law for magnetism (Figure 3.15) as follows:

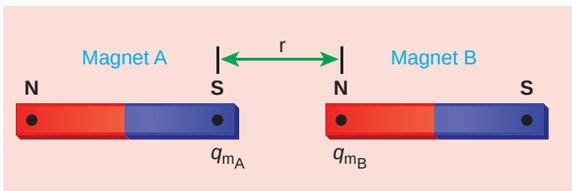


Figure 3.15 Coulomb's law – force between two magnetic pole strength

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